

Beyond PCA: A Gram-Schmidt Approach to Feature Extraction

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Abstract—Linear feature extraction at the presence of nonlinear dependencies among the data is a fundamental challenge in unsupervised learning. We propose using a Gram-Schmidt (GS) type orthogonalization process over function spaces in order to detect and remove redundant dimensions. Specifically, by applying the GS process over a family of functions which presumably captures the nonlinear dependencies in the data, we construct a series of covariance matrices that can either be used to identify new large-variance directions, or to remove those dependencies from the principal components. In the former case, we provide information-theoretic guarantees in terms of entropy reduction. In the latter, we prove that under certain assumptions the resulting algorithms detect and remove nonlinear dependencies whenever those dependencies lie in the linear span of the chosen function family. Both proposed methods extract linear features from the data while removing nonlinear redundancies. We provide simulation results on synthetic and real-world datasets which show improved performance over state-of-the-art feature extraction algorithms.

I. INTRODUCTION

The term *feature extraction* refers to a set of techniques used to reduce the number of features (or dimensions) in a dataset by mapping them to new features, while retaining some notion of informativeness. In the context of unsupervised learning, the goal is to simplify the data and make it easier to analyze or visualize, and to reduce the computational complexity and memory requirements of future analysis or communication algorithms [1].

In a sense, the ultimate goal of feature extraction is to identify and remove *redundancy*, that is, parts of the data which do not carry intrinsic information but rather can be determined by other parts of the data. It is well known that whenever such redundancy is linear, i.e., when certain features in the data are linear functions of other features, then Principal Component Analysis (PCA) [2] identifies and removes those redundant features. The paper at hand advances the state-of-the-art by presenting a new *linear* feature extraction technique which is capable of removing *nonlinear* redundancy from the data. If the redundancy is linear, our algorithm specifies to PCA. We focus on *unsupervised* feature extraction, i.e., that does not take into account any label information.

The well-known PCA technique works by analyzing the eigenvectors of the covariance matrix of the data (or an empirical approximation thereof), and extracts new features called *principal components*. These principal components are

uncorrelated and capture the variance in the original data in descending order. In contrast, we propose a family of Gram-Schmidt based algorithms which rely on eigenvector analysis of *alternative* covariance matrices, introduced next.

Suppose our data is sampled i.i.d from some unknown random variable $X = (X_1, \dots, X_d)^\top$ over $\mathbb{R}^d$. Our algorithms begin by fixing a finite family of linearly independent functions $\mathcal{F}(\mathbf{z})$ in some (non-random) variables $\mathbf{z} = (z_1, z_2, \dots)$ (e.g., low-degree polynomials). This $\mathcal{F}(\mathbf{z})$, which in practice should be chosen using domain expertise, presumably captures the anticipated dependencies between the features. Then, in an iterative process, the variables z_i of $\mathcal{F}(\mathbf{z})$ are substituted one-by-one with linear projections of X , i.e., of the form $X^\top \nu$ for some $\nu \in \mathbb{R}^d$, effectively turning those functions to random variables. Doing so enables to subtract those functions from X in a *Gram-Schmidt* (GS) process over function spaces (Section IV), thereby creating new data distributions with less redundancy. The covariance matrices of these new data distributions are the *alternative covariance matrices* mentioned earlier. In turn, eigen-analysis of these alternative covariance matrices enables to either define new high variance directions (Section V), or remove non-linear redundancy from the principal components (Section VI).

In more detail, the Gram-Schmidt process is a classic algebraic technique to convert a given basis of a subspace to an orthonormal basis of the same subspace, i.e., where all vectors are of length 1 and any two are orthogonal. We employ a well-known generalization of GS to function spaces, where vectors are replaced by random functions (i.e., functions which take random variables as inputs), and the inner product between two functions is the expectation of their product, computed according to the data distribution. Our algorithms apply for any finite set of functions, as long as they are square integrable according to the data distribution [3].

This GS process results in a new set of orthonormal random functions which enables to define new random variables d_j —one at each iteration j of the algorithm—by subtracting from X its projections on those orthonormal functions. This amounts to “nonlinear redundancy removal,” in the sense that these d_j ’s capture whatever remains in X should one set to zero the part of the redundancy which can be described by the orthonormal functions.

The $d_j(X)$ ’s give rise to alternative covariance matrices $\Sigma_{j+1} = \mathbb{E}[d_j(X)d_j(X)^\top]$. Standard eigenvector analysis over Σ_{j+1} provides insights into choosing the next vector ν_{j+1} , in order to substitute the next variable z_{j+1} of the functions in $\mathcal{F}(\mathbf{z})$ by a linear projection $Z_{j+1} = X^\top \nu_{j+1}$. This process stops once sufficiently many features have been

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extracted, or a maximum variance threshold has been crossed. This general framework, which is illustrated in Figure 1, is manifested in two complementing ways that are described next.

First, we wish to provide a simple and practical algorithm with theoretical guarantees. To this end, we present *Gram-Schmidt Functional Reduction* (GFR), a **linear** feature extraction technique that identifies new directions (different from the principal ones). We simply choose the directions ν_j as the eigenvectors which correspond to the largest eigenvalue (abbrv. largest eigenvector) of the respective Σ_j . We provide information-theoretic guarantees of bounded conditional entropy for the case of discrete data distributions (Section V).

Second, in order to gain deeper understanding of the manner in which our methods remove redundancy, we identify an ideal setting in which our approach is “perfect,” in the sense that $H(X|Z) = 0$. To this end, we present *Gram-Schmidt Component Analysis* (GCA), a **linear** feature extraction technique that identifies and removes nonlinear dependencies among the principal components. We formally prove that if these redundancies lie in the linear span of $\mathcal{F}$, and if they are *variance reducing* in a way that will be defined formally later, then they can be correctly identified and removed (Section VI).

Finally, in section VII, we apply the proposed algorithms to synthetic and real-world datasets and show their superior performance compared to state-of-the-art feature extraction algorithms. Section VIII concludes the paper.

II. PREVIOUS WORK

Unsupervised feature extraction has a vast literature and we summarize the main key contributions below. The most common feature extraction technique is Principal Component Analysis (PCA) [2] with multiple variants (such as generalized PCA [4], robust PCA [5], and kernel PCA [6]). Multidimensional scaling (MDS) is a class of methods whose objective is to maximize the scatter of the projection, under the rationale that doing so would yield the most informative projection [7]. Maximum Autocorrelation Factors (MAF) and Slow Feature Analysis are linear feature extraction methods for multivariate time series data [8], [9], hence address a problem different from the one considered herein. Locality Preserving Projections (LPP) aims to find a linear transformation of the data that preserves the local pairwise relationships between data points [10]. Independent Component Analysis (ICA) is a computational technique used in signal processing and data analysis. It is primarily used for separating a multivariate signal into additive, independent, and non-Gaussian source signals, but can also be employed for feature extraction [11]. Uniform Manifold Approximation and Projection (UMAP) is a nonlinear dimensionality reduction technique that is good at preserving both local and global structure of the data [12]. Autoencoders are a type of neural network designed for unsupervised learning, where the goal is to learn efficient coding of the input data [13].

Information theory also plays a role in feature extraction, since feature extraction can be seen as a special case of lossy

compression in information theory. More broadly, several previous works have taken an information-theoretic approach to feature extraction, such as [14]–[16], which studied the supervised setting and [17], which proposed to directly minimize the mutual information of the data and the extracted features heuristically using gradient-descent based methods.

In our experiments, we choose PCA, LPP, ICA (in its popular FastICA framework [18]), Kernel PCA, UMAP, and autoencoder as a baseline comparison of our methods.

III. NOTATIONS AND PRELIMINARIES

We denote vector random variables using capital letters X, Y, Z , etc., and in particular we let $X = (X_1, \dots, X_d)$ be the (unknown, zero mean) random variable over $\mathbb{R}^d$ (i.e., each X_i is over $\mathbb{R}$) from which the data is sampled i.i.d. We use $\mathbf{x} = (x_1, \dots, x_d), \mathbf{y} = (y_1, \dots, y_d)$, etc. to denote ordinary (that is, non-random) variables, and $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_d), \boldsymbol{\beta} = (\beta_1, \dots, \beta_d)$ etc. to denote vectors of scalars (i.e., $\alpha_i, \beta_i \in \mathbb{R}$). Further, for $\mathcal{S} \subseteq [d]$ (where $[d] \triangleq \{1, 2, \dots, d\}$), we use $X_{\mathcal{S}}$ (resp. $\mathbf{x}_{\mathcal{S}}$, etc.) to denote a vector of length $|\mathcal{S}|$ which contains the entries of X (resp. $\mathbf{x}$, etc.) that are indexed by $\mathcal{S}$.

We denote by $\mathcal{F}(\mathbf{z})$ a generic function family in the (ordinary) variables $z_1, \dots, z_d$, which are linearly independent over $\mathbb{R}$. For $\mathcal{S} \subseteq [d]$ we let $\mathcal{F}(\mathbf{z}_{\mathcal{S}})$ be the subset of $\mathcal{F}$ which contains all functions in $\mathcal{F}$ which depend only on a subset of the variables in $\mathbf{z}_{\mathcal{S}}$. Our Gram-Schmidt approach strongly relies on substituting the ordinary variables z_i by random ones Z_i , and hence for $\mathcal{S} = \{\sigma_1, \dots, \sigma_{|\mathcal{S}|}\} \subseteq [d]$ (with $\sigma_1 < \dots < \sigma_{|\mathcal{S}|}$) and a vector of random variables $Z = (Z_1, \dots, Z_d)$, we denote by $\mathcal{F}(Z_{\mathcal{S}})$ the result of substituting $z_i \leftarrow Z_{\sigma_i}$ for every $i \in [|\mathcal{S}|]$ in $\mathcal{F}(\mathbf{z}_{|\mathcal{S}|})$.

Example 1. Let $d = 3$ and $\mathcal{F}(\mathbf{z}) = \{z_1, z_2, z_3, z_1 z_2, z_1 z_2 z_3\}$. For $\mathcal{S} = [2]$ we have that $\mathcal{F}(\mathbf{z}_{\mathcal{S}}) = \{z_1, z_2, z_1 z_2\}$, and for $\mathcal{T} = \{1, 3\}$ we have $\mathcal{F}(Z_{\mathcal{T}}) = \{Z_1, Z_3, Z_1 Z_3\}$.

All function families $\mathcal{F}(\mathbf{z})$ we use in this paper contain all “singleton” functions, i.e., $\{f_i(\mathbf{z}) = z_i\}_{i=1}^d \subseteq \mathcal{F}(\mathbf{z})$. Notice that a set of the form $\mathcal{F}(Z_{\mathcal{S}})$ is a set of random function, and as such belongs to a Hilbert space in which the inner product between two functions f, g is $\mathbb{E}[fg]$. We note that all random variables in this paper are some functions of X , the random variable from which the data is sampled i.i.d, and all expectations are computed with respect to the distribution of X .

Remark 1. The random variables we use in this paper (including X) need not be known. We manipulate random variables as symbolic expressions, and only use them in the context of computing expectations. In practice, these expectations are approximated using empirical means. For example, we may manipulate X to obtain a function $\ell(X) = X_1 + X_2^2$, without the need to know how X is distributed. To avoid notational clutter, expressions of the form $\mathbb{E}[\ell(X)]$ are used in lieu of the empirical approximation $\mathbb{E}[\ell(X)] \approx \sum_{i=1}^N \ell(\boldsymbol{\alpha}_i) = \sum_{i=1}^N (\alpha_{i,1} + \alpha_{i,2}^2)$, where $\{\boldsymbol{\alpha}_i\}_{i=1}^N$ is the dataset at hand.

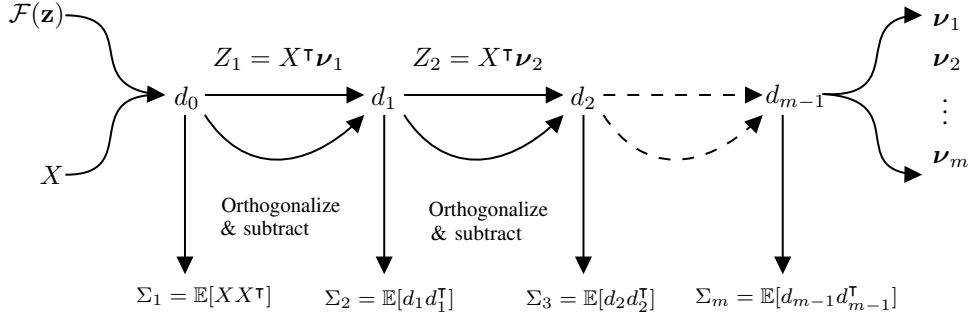


Fig. 1: A schematic description of the algorithms, Gram-Schmidt Functional Reduction (GFR, Section V) and Gram-Schmidt Component Analysis (GCA, Section VI). The computation of $\{\Sigma_i\}_{i=1}^m$ is done by iteratively subtracting orthogonalized functions, obtaining new random variables d_j 's, and the Σ_j 's are their covariance matrices. GFR and GCA vary in the way the ν_i 's are specified, either as the largest eigenvectors of the Σ_i 's (GFR), or as some principal directions (GCA).

As mentioned previously, our algorithms rely on the Gram-Schmidt process in a Hilbert space. To this end, we denote by $\hat{\mathcal{F}}(Z_S)$ the result of applying the Gram-Schmidt process over a function family $\mathcal{F}(Z_S)$; notice that $\text{span } \hat{\mathcal{F}}(Z_S) = \text{span } \mathcal{F}(Z_S)$. The full details of this Gram-Schmidt process are given in the following section. Finally, we use the algorithmic notation $a \leftarrow b$ to denote that the expression b is computed, and then assigned into the variable a .

IV. GRAM-SCHMIDT ORTHOGONALIZATION OVER FUNCTION SPACES

In this section we describe the core GS orthogonalization process in a function space on which all our algorithms are based. As mentioned earlier, we begin by fixing a function family $\mathcal{F}(\mathbf{z})$ in variables $\mathbf{z} = (z_1, \dots, z_d)$; other than being square integrable we do not make any particular assumptions on those functions, but in our experimental sections we focus on various polynomial functions.

Our algorithms (Sections V-VI) apply in an iterative manner, where in iteration j we choose some new feature Z_j to extract, substitute $z_j \leftarrow Z_j$ in the functions of $\mathcal{F}(\mathbf{z}_{[j]})$, and perform GS orthogonalization. To this end, Algorithm 1 operates at each iteration j of either of our algorithms by receiving an already orthogonalized $\hat{\mathcal{F}}(Z_1, \dots, Z_{j-1})$, new functions in $\mathcal{F}(\mathbf{z}_{[j]}) \setminus \mathcal{F}(\mathbf{z}_{[j-1]})$, and a new random variable Z_j . Then, for each member of $\mathcal{F}(\mathbf{z}_{[j]}) \setminus \mathcal{F}(\mathbf{z}_{[j-1]})$, the algorithm substitutes $z_i \leftarrow Z_i$ for every $i \in [j]$ (Line 6), subtracts its projection on already-orthogonalized functions (Line 7), normalizes (Line 8), and adds the result to $\mathcal{T}$ (Line 9). After the subtraction process we define a new (vector) random variable $d_j(X)$ from X (Line 12), whose covariance matrix is computed (Line 13) and returned as output.

V. GRAM-SCHMIDT FUNCTIONAL REDUCTION (GFR)

In Algorithm 2 below we propose *Gram-Schmidt Functional Reduction* (GFR), which uses the alternative covariance matrices Σ_j in order to extract new high variance directions. Similar to Section IV, we define a set of random variables $X = (X_1, \dots, X_d)^T$ to represent the data distribution (which need not be known, see Remark 1), and fix a function family $\mathcal{F}(\mathbf{z})$ in (non-random) variables $z_1, \dots, z_n$. In each

Algorithm 1: Orthogonalize($\hat{\mathcal{F}}(Z_1, \dots, Z_{j-1}), \mathcal{F}(\mathbf{z}_{[j]}) \setminus \mathcal{F}(\mathbf{z}_{[j-1]}), \{Z_i\}_{i=1}^j$)

1: **Input:**

- $\hat{\mathcal{F}}(Z_1, \dots, Z_{j-1})$: an orthogonalized function family in random variables $Z_1, \dots, Z_{j-1}$.
- $\mathcal{F}(\mathbf{z}_{[j]}) \setminus \mathcal{F}(\mathbf{z}_{[j-1]})$: all the functions in $\mathcal{F}(\mathbf{z})$ which depend on $\mathbf{z}_{[j]}$ but not only on $\mathbf{z}_{[j-1]}$.
- $\{Z_i\}_{i=1}^j$: the random variables $Z_1, \dots, Z_{j-1}$ which appear in $\hat{\mathcal{F}}(Z_1, \dots, Z_{j-1})$, and a new random variable Z_j .

2: **Output:** Orthogonalized function family $\hat{\mathcal{F}}(Z_1, \dots, Z_j)$, and a covariance matrix Σ_{j+1} .

3: **Initialize:** $\mathcal{T} \leftarrow \hat{\mathcal{F}}(Z_1, \dots, Z_{j-1})$.

4: Denote $\mathcal{F}(\mathbf{z}_{[j]}) \setminus \mathcal{F}(\mathbf{z}_{[j-1]}) \triangleq \{f_1(\mathbf{z}), \dots, f_\ell(\mathbf{z})\}$, where $f_1(\mathbf{z}) = z_j$.

5: **for** $k \leftarrow 1$ **to** ℓ **do**

6: $g(Z_1, \dots, Z_j) \leftarrow f_k(Z_1, \dots, Z_j)$
(create a random function g from a non-random one f_k by substituting $z_i \leftarrow Z_i$ for $i \in [j]$.)

7: $g(Z_1, \dots, Z_j) \leftarrow g(Z_1, \dots, Z_j) - \sum_{f \in \mathcal{T}} \mathbb{E}[gf]f$
(orthogonalize g with respect to $\mathcal{T}$.)

8: $g(Z_1, \dots, Z_j) \leftarrow \frac{g(Z_1, \dots, Z_j)}{\sqrt{\mathbb{E}[g(Z_1, \dots, Z_j)^2]}}$
(normalize g .)

9: $\mathcal{T} \leftarrow \mathcal{T} \cup \{g(Z_1, \dots, Z_j)\}$
(add g to $\mathcal{T}$)

10: **end for**

11: Define $\hat{\mathcal{F}}(Z_1, \dots, Z_j) = \mathcal{T}$.

12: Define $d_j(X) = X - \sum_{f \in \mathcal{T}} \mathbb{E}[Xf]f$.

13: Define $\Sigma_{j+1} = \mathbb{E}[d_j(X)d_j^T(X)]$.

14: Return $\Sigma_{j+1}, \mathcal{T}$.

iteration j , the algorithm begins by identifying the largest eigenvector ν_j of Σ_j (this would be the highest variance direction of $d_{j-1}(X)$, computed in Line 12 of Algorithm 1). If the variance of this direction is less than some threshold ϵ^2 , the algorithm stops, and otherwise it continues by specifying the next random variable Z_j as $Z_j = X^T \nu_j$, which enables

the next call to “Orthogonalize.” Again, the variables X , and their functions, are merely symbolic expressions. They are only required in the context of their expectation, which in reality is approximated using empirical means.

Algorithm 2: Gram-Schmidt Functional Reduction (GFR)

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1: Input: A function family  $\mathcal{F}(\mathbf{z})$  and a threshold  $\epsilon > 0$ .
2: Output: Extracted directions  $\nu_1, \dots, \nu_m$  ( $m$  is a
   varying number that depends on  $\mathcal{F}(\mathbf{z})$ ,  $\epsilon$ , and  $X$ ).
3: Initialize:  $\Sigma_1 = \mathbb{E}[X X^\top]$ .
4: for  $j \leftarrow 1$  to  $d$  do
5:   Let  $\nu_j$  be the largest unit-norm eigenvector of  $\Sigma_j$ ,
     i.e.,  $\nu_j = \arg \max_{\|\nu\|=1} \nu^\top \Sigma_j \nu$ .
6:   if  $\nu_j^\top \Sigma_j \nu_j \leq \epsilon^2$  then
7:     break.
8:   else
9:     Define  $Z_j = X^\top \nu_j$ .
10:  end if
11:   $\Sigma_{j+1}, \hat{\mathcal{F}}(Z_1, \dots, Z_j) =$ 
    Orthogonalize( $\hat{\mathcal{F}}(Z_1, \dots, Z_{j-1}), \mathcal{F}(\mathbf{z}_{[j]}) \setminus$ 
     $\mathcal{F}(\mathbf{z}_{[j-1]}), \{Z_i\}_{i=1}^j$ ).
12: end for

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Next, we state the information-theoretic guarantees of GFR. While GFR shows significant gains over state-of-the-art linear (and some nonlinear) methods with real-world datasets, its information-theoretic guarantees require the data distribution X to be over a finite alphabet. In our experiments it is shown that setting $\mathcal{F}(\mathbf{z})$ to be multilinear polynomials of low degree reduces the residual variance¹ in real-world datasets by up to an order of magnitude in comparison with PCA.

Theorem 1. *Let X be over a discrete domain $\mathcal{X}^d$, for some $\mathcal{X} \subseteq \mathbb{R}$, and let $Z = (Z_1, \dots, Z_m) = (X^\top \nu_1, \dots, X^\top \nu_m)$, where $\nu_1, \dots, \nu_m$ is the output of GFR whose input is a given ϵ and $\mathcal{F}(\mathbf{z})$. Then $H(X|Z) \leq dO(\epsilon)$.*

Before proving Theorem 1, the following technical statement is in place. This statement essentially shows that once the algorithm stops, the variance of the data in *all potential subsequent directions* is small. Therefore, we let $\nu_{m+1}, \dots, \nu_d$ be any completion of the extracted directions $\nu_1, \dots, \nu_m$ to an orthonormal basis (the fact that $\nu_1, \dots, \nu_m$ are orthonormal will be proved in Lemma 2), and let $\Sigma_{m+1}, \dots, \Sigma_d$ be the covariance matrices which follow by completing the orthogonalization process to produce $\hat{\mathcal{F}}(Z_1, \dots, Z_d)$, done throughout GFR, while fixing the random variables $Z_j = X^\top \nu_j$ for all $j \in \{m+1, \dots, d\}$.

Lemma 1. *The matrices $\{\Sigma_j\}_{m+1}^d$ that are defined during the completion of the orthogonalization of $\mathcal{F}(\mathbf{z})$ satisfy that $\nu_j^\top \Sigma_j \nu_j \leq \epsilon^2$ for all $j \in \{m+1, \dots, d\}$.*

¹I.e., the maximum amount of variance that is left in the data distribution after extracting said features. This is captured in PCA by the $(m+1)$ 'th largest eigenvalue of the covariance matrix when extracting m features.

The following technical statement is required for the subsequent proof of Lemma 1.

Lemma 2. *In GFR, we have the following.*

- (a) *For $i < j$ in $[d]$ we have $\Sigma_j = \Sigma_i - \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_{i-1}} \mathbb{E}[fX] \mathbb{E}[X^\top f]$.*
- (b) *The vectors $\nu_1, \dots, \nu_m$ are orthonormal.*

Proof of Lemma 2.(a). It is readily verified that $d_{j-1}(X)$ can be written as

$$d_{j-1}(X) = d_{i-1}(X) - \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Xf]f, \quad (1)$$

and that $d_{i-1}(X)$ can be written as

$$d_{i-1}(X) = X - \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Xf]f. \quad (2)$$

Hence, it follows from (1) that

$$\begin{aligned}
\Sigma_j &= \mathbb{E}[d_{j-1}(X)d_{j-1}(X)^\top] \\
&= \mathbb{E} \left[\left(d_{i-1}(X) - \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Xf]f \right) \right. \\
&\quad \left. \left(d_{i-1}(X) - \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Xf]f \right)^\top \right] \\
&= \Sigma_i - \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_{i-1}} \mathbb{E}[f d_{i-1}(X)] \mathbb{E}[X^\top f] \\
&\quad - \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_{i-1}} \mathbb{E}[fX] \mathbb{E}[d_{i-1}(X)^\top f] \\
&\quad + \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Xf] \mathbb{E}[X^\top f], \quad (3)
\end{aligned}$$

where the last summand follows from the orthonormality of $\hat{\mathcal{F}}_{j-1}$. Further, it follows from (2) that every $f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_{i-1}$ satisfies

$$\mathbb{E}[f d_{i-1}(X)] = \mathbb{E}[f \cdot (X - \sum_{g \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Xg]g)] = \mathbb{E}[fX],$$

and therefore (3) = $\Sigma_i - \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Xf] \mathbb{E}[X^\top f]$ as required. $\square$

Proof of Lemma 2.(b). By definition, we have that ν_j is the largest eigenvector of Σ_j for every $j \in [m]$, and let $\lambda_j \neq 0$ be the respective eigenvalue (if $\lambda_j = 0$ the algorithm stops). We show that for any $j \in [m]$, ν_j is orthogonal to $\nu_1, \dots, \nu_{j-1}$, as follows.

Let $i < j$, and observe that $\nu_i^\top \Sigma_j \nu_j = \lambda_j \nu_i^\top \nu_j$. In addition, it follows from part (a) that

$$\Sigma_j = \Sigma_i - \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_{i-1}} \mathbb{E}[fX] \mathbb{E}[X^\top f],$$

and hence

$$\begin{aligned}
\nu_i^\top \Sigma_j \nu_j &= \nu_i^\top \left(\Sigma_i - \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_{i-1}} \mathbb{E}[fX] \mathbb{E}[X^\top f] \right) \nu_j \\
&= \lambda_i \nu_i^\top \nu_j - \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_{i-1}} \mathbb{E}[f \cdot \nu_i^\top X] \mathbb{E}[X^\top \nu_j \cdot f]. \quad (4)
\end{aligned}$$

For $i \in [m]$, let $\tilde{Z}_i$ and $\hat{Z}_i$ be the orthogonalized and orthonormalized version of Z_i , respectively. By definition we have that $\nu_i^\top X = Z_i = \tilde{Z}_i + \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Z_i f]f$, and hence every $f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_{i-1}$ satisfies

$$\begin{aligned}
\mathbb{E}[f \cdot \nu_i^\top X] &= \mathbb{E}[f \cdot (\tilde{Z}_i + \sum_{g \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Z_i g]g)] \\
&= \mathbb{E}[f \tilde{Z}_i] = \begin{cases} \|\tilde{Z}_i\| & \text{if } f = \hat{Z}_i \\ 0 & \text{otherwise.} \end{cases} \quad (5)
\end{aligned}$$

Therefore,

$$\begin{aligned}
(4) &= \lambda_i \boldsymbol{\nu}_i^\top \boldsymbol{\nu}_j - \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_{i-1}} \mathbb{E}[f \tilde{Z}_i] \mathbb{E}[X^\top \boldsymbol{\nu}_j \cdot f] \\
&= \lambda_i \boldsymbol{\nu}_i^\top \boldsymbol{\nu}_j - \|\tilde{Z}_i\| \mathbb{E}[X^\top \boldsymbol{\nu}_j \cdot \tilde{Z}_i] = (\lambda_i \boldsymbol{\nu}_i^\top - \mathbb{E}[X^\top \tilde{Z}_i]) \boldsymbol{\nu}_j.
\end{aligned} \tag{6}$$

Now observe that

$$\begin{aligned}
\mathbb{E}[X \tilde{Z}_i] &= \mathbb{E}[X(Z_i - \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Z_i f] f)] \\
&= \mathbb{E}[X(\boldsymbol{\nu}_i^\top X - \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[\boldsymbol{\nu}_i^\top X f] f)] \\
&= \mathbb{E}[(X X^\top) \boldsymbol{\nu}_i] - \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[X f] \mathbb{E}[X^\top \boldsymbol{\nu}_i f] \\
&= (\mathbb{E}[X X^\top] - \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[f X] \mathbb{E}[X^\top f]) \boldsymbol{\nu}_i \\
&\stackrel{(\star)}{=} \Sigma_i \boldsymbol{\nu}_i = \lambda_i \boldsymbol{\nu}_i,
\end{aligned}$$

where $(\star)$ follows by applying part (a) on Σ_i and Σ_1 , and hence (6) = 0. Subsequently, we have $\lambda_j \boldsymbol{\nu}_j^\top \boldsymbol{\nu}_j = \boldsymbol{\nu}_j^\top \Sigma_j \boldsymbol{\nu}_j = 0$, which implies that $\boldsymbol{\nu}_j$ is orthogonal to $\boldsymbol{\nu}_i$ for all $i \in [j-1]$, and the claim follows. $\square$

Proof of Lemma 1. According to the stopping criterion in GFR, it follows that

$$\begin{aligned}
&\boldsymbol{\nu}_j^\top \Sigma_j \boldsymbol{\nu}_j > \epsilon^2 \text{ for all } j \in [m] \\
\max_{\|\mathbf{v}\|=1} \mathbf{v}^\top \Sigma_{m+1} \mathbf{v} &\leq \epsilon^2.
\end{aligned} \tag{7}$$

Assume for contradiction that there exists $j \in \{m+1, \dots, d\}$ such that $\boldsymbol{\nu}_j^\top \Sigma_j \boldsymbol{\nu}_j > \epsilon^2$, and observe that $j > m+1$, since otherwise the existence of $\boldsymbol{\nu}_{m+1}$ contradicts (7). It follows from Lemma 2.(a) that

$$\Sigma_j = \Sigma_{m+1} - \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_m} \mathbb{E}[f X] \mathbb{E}[X^\top f],$$

and therefore

$$\begin{aligned}
\boldsymbol{\nu}_j^\top \Sigma_j \boldsymbol{\nu}_j &= \boldsymbol{\nu}_j^\top \Sigma_{m+1} \boldsymbol{\nu}_j - \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_m} \mathbb{E}[f \cdot \boldsymbol{\nu}_j^\top X]^2 \\
&\stackrel{(7)}{\leq} \epsilon^2 - \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_m} \mathbb{E}[f \cdot \boldsymbol{\nu}_j^\top X]^2.
\end{aligned}$$

Therefore, since $\boldsymbol{\nu}_j^\top \Sigma_j \boldsymbol{\nu}_j > \epsilon^2$, it follows that

$$- \sum_{f \in \hat{\mathcal{F}}_{j-1} \setminus \hat{\mathcal{F}}_m} \mathbb{E}[f \cdot \boldsymbol{\nu}_j^\top X]^2 > 0,$$

which is a contradiction since $\mathbb{E}[f \cdot \boldsymbol{\nu}_j^\top X]^2 \geq 0$ for every f . $\square$

Proof of Theorem 1. Let $\boldsymbol{\nu}_1, \dots, \boldsymbol{\nu}_m$ be the vectors found throughout GFR, and let $\boldsymbol{\nu}_{m+1}, \dots, \boldsymbol{\nu}_d$ be a completion of $\boldsymbol{\nu}_1, \dots, \boldsymbol{\nu}_m$ to an orthonormal basis of $\mathbb{R}^d$. Further, as in Lemma 1, let $\hat{\mathcal{F}}(Z_1, \dots, Z_d)$ be the result of completing the orthogonalization process, done throughout GFR, to orthogonalization of $\mathcal{F}(\mathbf{z})$ in its entirety, where the variables $Z^\perp \triangleq (Z_{m+1}, \dots, Z_d)$ are specified as $Z_j = X^\top \boldsymbol{\nu}_j$ for all $j \in \{m+1, \dots, d\}$.

Since the matrix whose rows are the $\boldsymbol{\nu}_i$'s has determinant 1, we have:

$$\begin{aligned}
H(X|Z) &= H(\{X^\top \boldsymbol{\nu}_j\}_{j=1}^d | \{X^\top \boldsymbol{\nu}_j\}_{j=1}^m) \\
&= H(\{X^\top \boldsymbol{\nu}_j\}_{j=m+1}^d | \{X^\top \boldsymbol{\nu}_j\}_{j=1}^m) \\
&= H(Z^\perp | Z) = \sum_{i=m+1}^d H(Z_i | (Z_j)_{j=1}^{i-1}), \tag{8}
\end{aligned}$$

where the last equality follows from the chain rule for information entropy. Since

$$\tilde{Z}_i = Z_i - \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Z_i f] f, \tag{9}$$

it follows that

$$\begin{aligned}
H(Z_i | (Z_j)_{j=1}^{i-1}) &= H(\tilde{Z}_i + \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Z_i f] f | (Z_j)_{j=1}^{i-1}) \\
&\stackrel{(a)}{=} H(\tilde{Z}_i | (Z_j)_{j=1}^{i-1}) \stackrel{(b)}{\leq} H(\tilde{Z}_i), \tag{10}
\end{aligned}$$

where (a) follows since the expression $\sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Z_i f] f$ is uniquely determined by the variables $Z_1, \dots, Z_{i-1}$ (every $f \in \hat{\mathcal{F}}_{i-1}$ is a deterministic function of $Z_1, \dots, Z_{i-1}$, and the coefficients $\mathbb{E}[Z_i f]$ are constants), and (b) follows since conditioning reduces entropy. Combining (8) with (10) we have that $H(X|Z) \leq \sum_{i=m+1}^d H(\tilde{Z}_i)$, and hence it remains to bound $H(\tilde{Z}_i)$ for all $i \in \{m+1, \dots, d\}$.

To this end let $i \in \{m+1, \dots, d\}$, define $\alpha_i \triangleq \min\{|\tilde{Z}_i(\boldsymbol{\alpha})| : \boldsymbol{\alpha} \in \mathcal{X}^d, \tilde{Z}_i(\boldsymbol{\alpha}) \neq 0\}$ and $\alpha_{\min} = \min\{\alpha_i\}_{i=m+1}^d$, and observe that by Markov's inequality we have

$$\begin{aligned}
\Pr(\tilde{Z}_i(X) \neq 0) &= \Pr(|\tilde{Z}_i(X)| \geq \alpha_i) = \Pr(\tilde{Z}_i(X)^2 \geq \alpha_i^2) \\
&\leq \frac{\mathbb{E}[\tilde{Z}_i^2]}{\alpha_i^2} \leq \frac{\mathbb{E}[\tilde{Z}_i^2]}{\alpha_{\min}^2}.
\end{aligned} \tag{11}$$

Moreover, recall that

$$\begin{aligned}
d_{i-1}(X) &= X - \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[X f] f, \text{ and} \\
\Sigma_i &= \mathbb{E}[d_{i-1}(X) d_{i-1}(X)^\top],
\end{aligned}$$

and therefore

$$\begin{aligned}
\boldsymbol{\nu}_i^\top \Sigma_i \boldsymbol{\nu}_i &= \boldsymbol{\nu}_i^\top \mathbb{E}[d_{i-1}(X) d_{i-1}(X)^\top] \boldsymbol{\nu}_i \\
&= \boldsymbol{\nu}_i^\top \mathbb{E}[(X - \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[X f] f)(X - \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[X f] f)^\top] \boldsymbol{\nu}_i \\
&= \mathbb{E}[(\boldsymbol{\nu}_i^\top X - \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[\boldsymbol{\nu}_i^\top X f] f)(X^\top \boldsymbol{\nu}_i - \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[X^\top \boldsymbol{\nu}_i f] f)] \\
&= \mathbb{E}[(Z_i - \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Z_i f] f)(Z_i - \sum_{f \in \hat{\mathcal{F}}_{i-1}} \mathbb{E}[Z_i f] f)] \stackrel{(9)}{=} \mathbb{E}[\tilde{Z}_i^2].
\end{aligned}$$

Consequently, it follows that

$$(11) = \frac{\boldsymbol{\nu}_i^\top \Sigma_i \boldsymbol{\nu}_i}{\alpha_{\min}^2} \stackrel{\text{Lemma 1}}{\leq} \frac{\epsilon^2}{\alpha_{\min}^2}.$$

Now, by using the grouping rule [19, Ex. 2.27] we have

$$H(\tilde{Z}_i) \leq h_b(\epsilon^2/\alpha_{\min}^2) + \frac{\epsilon^2}{\alpha_{\min}^2} \log |\mathcal{X}|,$$

where h_b is the binary entropy function. Finally, using the bound $h_b(p) \leq 2\sqrt{p(1-p)} \leq 2\sqrt{p}$, it follows that

$$\begin{aligned}
H(X|Z) &\leq \sum_{i=m+1}^d H(\tilde{Z}_i) \\
&\leq (d-m)(h_b(\epsilon^2/\alpha_{\min}^2) + (\epsilon^2/\alpha_{\min}^2) \log |\mathcal{X}|) \\
&\leq (d-m)(2\epsilon/\alpha_{\min} + (\epsilon/\alpha_{\min}) \log |\mathcal{X}|) = dO(\epsilon). \square
\end{aligned}$$

VI. GRAM-SCHMIDT COMPONENT ANALYSIS (GCA)

In this section we present *Gram-Schmidt Component Analysis* (GCA) in Algorithm 3, and formally prove that it correctly removes nonlinear redundancy in an ideal setting. Recall that $X = (X_1, \dots, X_d)^\top$ is the random variable from which the data is sampled (which need not be known, see

Remark 1), and let $\rho_1, \dots, \rho_d$ be the principal directions of X , not necessarily ordered by eigenvalue magnitude. In this ideal setting, we assume that some $n \leq d$ principal components of X are determined by random variables $Y_1, \dots, Y_n$; we call those *non-redundant principal components*. The remaining $d - n$ components, which we call *redundant principal components*, are given by (potentially nonlinear) functions $h_{n+1}, \dots, h_d$ of $Y_1, \dots, Y_n$ which lie in the linear span of the function family $\mathcal{F}(\mathbf{z})$. In Theorem 2 below we provide conditions by which in such ideal setting, GCA correctly identifies the non-redundant principal components, effectively removing the nonlinear redundancies $h_{n+1}, \dots, h_d$.

In each iteration j , the algorithm begins by identifying the principal directions ρ_i of X in which the random variable d_j (see Algorithm 1) has no variance, i.e., $\rho_i^\top \Sigma_j \rho_i = 0$ (or smaller than some positive δ in the empirical variant). These principal components are then subtracted from X to produce $\bar{X}_j$, and the most variant component ρ_{ℓ_j} of $\bar{X}_j$ is identified via the covariance matrix of $\bar{X}_j$. By specifying Z_j to a projection of X on ρ_{ℓ_j} , the next call to “Orthogonalize” can be made.

Algorithm 3: Gram-Schmidt Component Analysis (GCA)

- 1: **Input:** A function family $\mathcal{F}(\mathbf{z})$ and a threshold $\epsilon > 0$.
 - 2: **Output:** Extracted directions $\nu_1, \dots, \nu_n$.
 - 3: **Initialize:** $\Sigma_1 = \mathbb{E}[XX^\top]$, and let $\{\rho_i\}_{i=1}^d$ be the normalized principal components of X .
 - 4: **for** $j \leftarrow 1$ **to** n **do**
 - 5: Let $\mathcal{E}_j = \{i | \rho_i^\top \Sigma_j \rho_i = 0\}$ (or $\rho_i^\top \Sigma_j \rho_i < \delta$ for some small δ in the empirical variant).
 - 6: Define $\bar{X}_j = X - \sum_{i \in \mathcal{E}_j} X^\top \rho_i \cdot \rho_i$.
 - 7: Let $\ell_j = \arg \max_{i \in [d] \setminus \mathcal{E}_j} \rho_i^\top \mathbb{E}[\bar{X}_j \bar{X}_j^\top] \rho_i$ (breaking ties arbitrarily).
 - 8: Define $\nu_j = \rho_{\ell_j}$ and $Z_j = X^\top \nu_j$.
 - 9: $\Sigma_{j+1}, \hat{\mathcal{F}}(Z_1, \dots, Z_j) =$
 Orthogonalize($\hat{\mathcal{F}}(Z_1, \dots, Z_{j-1}), \mathcal{F}(\mathbf{z}_{[j]}) \setminus \mathcal{F}(\mathbf{z}_{[j-1]}), \{Z_i\}_{i=1}^j$).
 - 10: **end for**
-

We now prove that if the redundancies have less variance than their constituent variables, then GCA correctly identifies and removes the redundant components in the data.

Theorem 2. *For positive integers $d \geq n$, where n is the number of non-redundant features, let $\mathcal{F}(\mathbf{z})$ be a set of functions in (non-random) variables $\mathbf{z} = (z_1, \dots, z_n)$, let $h_1, \dots, h_d \in \text{span } \mathcal{F}(\mathbf{z})$ with $h_i(\mathbf{z}) = z_i$ for $i \in [n]$, and let $Y = (Y_1, \dots, Y_n)$ be random variables such that for every $j \in \{n+1, \dots, d\}$,*

$$\text{Var}(h_j(Y)) < \min_{\{i | h_j \text{ depends on } Y_i\}} \text{Var}(Y_i). \quad (12)$$

For a random variable X in $\mathbb{R}^d$ of the form $X = \sum_{i=1}^d \rho_i h_i(Y)$, where the ρ_i 's are the principal directions (and hence the h_i are orthogonal) of X , GCA with input $\mathcal{F}(\mathbf{z})$ outputs $\{\nu_i\}_{i=1}^n = \{\rho_i\}_{i=1}^n$ such that $\{Z_i\}_{i=1}^n =$

$\{X^\top \nu_i\}_{i=1}^n = \{h_i(Y)\}_{i=1}^n = \{Y_i\}_{i=1}^n$, i.e., it identifies the non-redundant principal directions correctly, and therefore $H(X|Z) = 0$.

Proof of Theorem 2. The claim is proved by induction on the iteration index j in GCA. For the base case $j = 1$, observe that $\mathcal{E}_1 = \emptyset$ and $\bar{X}_1 = X$, and hence ν_1 is the maximizer of $\rho_i^\top \mathbb{E}[XX^\top] \rho_i$ over $i \in [d]$. Since $X = \sum_{i=1}^d \rho_i h_i(Y)$ with the ρ_i 's orthonormal, it follows that

$$\rho_i^\top \mathbb{E}[XX^\top] \rho_i = \mathbb{E}[h_i(Y)^2] = \text{Var}(h_i(Y)), \quad (13)$$

where the last equality follows since X is centered. Hence, the maximizing vector ν_1 of $\rho_i^\top \mathbb{E}[XX^\top] \rho_i$ over $i \in [d]$, according to (12), is one of $\rho_1, \dots, \rho_n$ (i.e., the non-redundant principal directions), say, ρ_{ℓ_1} for some $\ell_1 \in [n]$. This implies that $Z_1 = X^\top \rho_{\ell_1} = Y_{\ell_1}$, i.e., that the first non-redundant principal direction is identified correctly.

Now, the induction hypothesis is that $\nu_i = \rho_{\ell_i}$ for $i \in [j-1]$ for some $j \geq 2$ and some distinct $\ell_1, \dots, \ell_{j-1} \in [n]$, i.e., that $Z_i = X^\top \rho_{\ell_i} = Y_{\ell_i}$. We wish to show that $\nu_j = \rho_{\ell_j}$ for some $\ell_j \in [n] \setminus \{\ell_1, \dots, \ell_{j-1}\}$, i.e., that the next extracted direction is a non-redundant principal direction that was not extracted so far. Since $X = \sum_{i=1}^d \rho_i h_i(Y)$, then $d_{j-1}(X)$ can be written as

$$\begin{aligned} d_{j-1}(X) &= X - \sum_{f \in \hat{\mathcal{F}}_{j-1}} \mathbb{E}[X f(Z_1, \dots, Z_{j-1})] f(Z_1, \dots, Z_{j-1}) \\ &= X - \sum_{f \in \hat{\mathcal{F}}_{j-1}} \mathbb{E}[(\sum_{i=1}^d \rho_i h_i(Y)) f(Y_{\ell_1}, \dots, Y_{\ell_{j-1}})] \\ &\quad f(Y_{\ell_1}, \dots, Y_{\ell_{j-1}}). \end{aligned} \quad (14)$$

For $i \in [d]$ let m_i be the smallest integer such that $h_i(\mathbf{z}) \in \text{span}\{f_1(\mathbf{z}), \dots, f_{m_i}(\mathbf{z})\}$. After substituting the ordinary variables z_i by random ones Z_i and orthonormalizing $f_1, \dots, f_{m_i}$, we have $h_i(Y) = \sum_{a=1}^{m_i} \mathbb{E}[h_i(Y) f_a] f_a$ for every i since the f_a 's are orthonormal. Therefore,

$$d_{j-1}(X) = X - \sum_{f \in \hat{\mathcal{F}}_{j-1}} \mathbb{E}[(\sum_{i=1}^d \rho_i (\sum_{a=1}^{m_i} \mathbb{E}[h_i(Y) f_a] f_a)) f] f.$$

By denoting $\hat{\mathcal{F}}_{j-1} = \{f_k\}_{k=1}^p$ we have

$$\begin{aligned} d_{j-1}(X) &= X - \sum_{k=1}^p \mathbb{E}[(\sum_{i=1}^d \rho_i (\sum_{a=1}^{m_i} \mathbb{E}[h_i(Y) f_a] f_a)) f_k] f_k \\ &= X - \sum_{k=1}^p \sum_{i=1}^d \sum_{a=1}^{m_i} \rho_i \mathbb{E}[h_i(Y) f_a] \mathbb{E}[f_k f_a] f_k \\ &= X - \sum_{i=1}^d \sum_{a=1}^{\min\{p, m_i\}} \rho_i \mathbb{E}[h_i(Y) f_a] f_a \\ &= \sum_{i=1}^d \rho_i \left(h_i(Y) - \sum_{a=1}^{\min\{p, m_i\}} \mathbb{E}[h_i(Y) f_a] f_a \right). \end{aligned} \quad (15)$$

Clearly, whenever $p > m_i$, the expression in the parentheses in (15) equals zero, i.e., the i th component ρ_i of d_{j-1} , as well as its variance, are zero. The definition of $\bar{X}_j$ using the set $\mathcal{E}_j$, and the induction hypothesis, imply that $\bar{X}_j$ is equal to X minus all $\rho_i h_i$ such that h_i depends on a subset of the variables $Z_1 = Y_{\ell_1}, \dots, Z_{j-1} = Y_{\ell_{j-1}}$. By the definition of X , the remaining components of $\bar{X}_j$ are either of the form $\rho_i Y_i$ for $i \in [n] \setminus \{\ell_1, \dots, \ell_{j-1}\}$, or of the form $\rho_i h_i(Y)$ for $i \in \{n+1, \dots, d\}$, where h_i does not depend exclusively on $\{Z_1, \dots, Z_{j-1}\}$. Therefore, since each such h_i depends on at least one remaining variable Y_a

²We assume without loss of generality that the principal components have positive variance.

for $a \in [n] \setminus \{\ell_1, \dots, \ell_{j-1}\}$, Eqs. (12) and (13) imply that in iteration j , the maximizer ρ_{ℓ_j} of $\rho_i^\top \mathbb{E}[\bar{X}_j \bar{X}_j^\top] \rho_i$ over $i \in [d] \setminus \mathcal{E}_j$ is some $\ell_j \in [n] \setminus \{\ell_1, \dots, \ell_{j-1}\}$, as required. Completing n steps of this induction implies that $\{\nu_i\}_{i=1}^n = \{\rho_{\ell_i}\}_{i=1}^n = \{\rho_i\}_{i=1}^n$, which concludes the proof. $\square$

Example 2. For $d = 4$ and $n = 3$, let $X = \rho_1 Y_1 + \rho_2 Y_2 + \rho_3 Y_3 + \rho_4 Y_1 Y_2$ for some orthonormal $\rho_i \in \mathbb{R}^d$, where Y_1, Y_2, Y_3 are mutually independent zero-mean Gaussian with variances 0.9, 0.8, 0.7, respectively. Since $\text{Var}(Y_1 Y_2) = 0.72 < \min\{\text{Var}(Y_1), \text{Var}(Y_2)\}$, it follows that Eq. (12) holds, and GCA will correctly extract Y_1 , then Y_2 , skip $Y_1 Y_2$, and extract Y_3 ; knowing Y_1, Y_2, Y_3 determines this data distribution completely, hence $H(X|Z) = 0$. In contrast, PCA for extracting 3 features will extract $Y_1, Y_2, Y_1 Y_2$ and discard Y_3 , resulting in nonzero conditional entropy.

VII. EXPERIMENTAL RESULTS

In this section, the theoretical results and performance of the proposed algorithms are evaluated through simulation studies over synthetic and real-world datasets, where expectations are approximated using empirical means. In section VII-A, we support the claim that GFR significantly reduces the data's residual variance in comparison with PCA. Intuitively, residual variance captures how variant the data is after extracting features. Formally, in PCA residual variance after extracting m features is captured by the $(m+1)$ 'th eigenvalue of the covariance matrix Σ_{m+1} , and in GFR it is the largest eigenvalue of the alternative covariance matrix Σ_{m+1} , i.e., the parameter ϵ^2 (see Algorithm 2). In section VII-B, we evaluate GFR for classification tasks over benchmark datasets and compare the results to other well-known unsupervised feature extraction algorithms, like Kernel PCA, FastICA, UMAP, autoencoders, and LPP. We use either one of the methods for feature extraction while disregarding the labels. The labels are later re-attached, a classifier is trained over the extracted features, and the resulting accuracies are compared. In section VII-C, we study the performance of GCA. Although its correctness is formally proven, we verify the effect of employing empirical covariance matrices and the size of the datasets on GCA's performance.

A. Experiments for Residual Variance Reduction of GFR

To support our claim that GFR results in significant variance reduction in comparison to PCA, we apply GFR to benchmark datasets taken from UCI repository [20]. The properties of the tested benchmark datasets are provided in Table I. In all the experiments, features were standardized before applying any feature extraction algorithm.

Dataset	MNIST	USPS	COIL-20
Features	784	256	1024
Samples	60000	7291	1440

TABLE I: Properties of the tested benchmark datasets.

We apply PCA and GFR, where $\mathcal{F}(\mathbf{z})$ in GFR is set as multilinear polynomials of degrees up to 2, 3, or 4 to the

benchmark datasets, and the results are shown in Table II. Note that GFR with multilinear polynomials of degree 1 is PCA. Table II shows the number of features that must be extracted to achieve residual variance of ϵ^2 , in PCA vs. GFR with multilinear polynomials of degrees up to 2, 3, and 4. As demonstrated in the experimental results, by increasing the degree of multilinear polynomials, GFR extracts fewer features given the same threshold ϵ^2 . Table II clearly shows a drastic reduction of residual variance as the degree of the multilinear polynomials increases, showcasing that the redundancies in these datasets are well described by them.

MNIST Dataset						
Degree of Polynomial	Residual variance					
	0.01	0.05	0.1	0.2	0.5	0.75
1 (PCA)	696	576	451	344	233	191
2	205	145	114	84	53	43
3	55	49	45	39	31	26
4	29	28	27	25	22	20

USPS Dataset					
Degree of Polynomial	Residual variance				
	0.1	0.15	0.2	0.5	1
1 (PCA)	208	130	113	70	45
2	61	52	47	30	21
3	29	27	25	20	16
4	20	19	19	17	14

COIL-20 Dataset						
Degree of Polynomial	Residual variance					
	0.01	0.02	0.05	0.1	0.2	0.5
1 (PCA)	577	477	353	264	189	120
2	50	47	42	38	33	27
3	21	20	19	18	17	15
4	14	14	14	14	13	12

TABLE II: Residual variance of GFR in benchmark datasets.

B. Experiments for Classification Accuracy of GFR

In this subsection, the experiments are validating GFR for classification tasks on the benchmark datasets. In Table III, we apply GFR with multilinear polynomials of degree 4, against Kernel PCA, FastICA, UMAP, Autoencoder, and LPP. Since UMAP and autoencoder have hyperparameters to tune, we have tuned the parameters and reported the highest accuracy we obtained in the experiments. The classification accuracy in the benchmark datasets shows overall superior performance of GFR in comparison to other algorithms. We used a support vector machine classifier with radial basis function as kernel. 5-fold cross-validation on the entire datasets is used to validate the performance of the algorithms.

C. Experimental Results for GCA

We evaluate GCA on the basis of its success in eliminating all redundant features among the principal components in synthetic datasets. Similar to Section VI, suppose $X = \sum_{i=1}^d \rho_i h_i(Y)$. In these experiments, we use $d = 30$ features, among them, $n = 15$ are non-redundant. We let $h_1 = Y_1, \dots, h_n = Y_n$ be mutually independent non-redundant principal components, chosen from $\mathcal{N}(0, \text{Unif}(0, 15))$. We define redundancies using two types of multilinear polynomials up to degree 2 or 3 and choose $(d - n)$ redundant components, $h_{n+1}, \dots, h_d$, randomly of the form

MNIST Dataset						
Method	Number of Extracted Features					
	29	28	27	25	22	20
GFR	96.91	96.90	96.87	96.79	96.63	96.46
Kernel PCA	93.11	93.02	92.87	92.72	92.69	92.58
FastICA	95.01	94.98	94.94	94.85	94.52	94.35
UMAP	96.34	96.27	96.19	96.14	96.11	96.09
Autoencoder	96.75	96.71	96.71	96.65	96.56	96.33
LPP	87.13	86.95	86.8	86.31	86.03	85.94

USPS Dataset				
Method	Number of Extracted Features			
	20	19	17	14
GFR	97.02	97	96.86	96.04
Kernel PCA	96.90	96.69	96.52	95.86
FastICA	96.14	95.77	95.61	95.54
UMAP	96.24	96.28	96.27	96.31
Autoencoder	97.01	96.91	96.74	96.21
LPP	93.54	93.48	93.29	92.32

COIL-20 Dataset			
Method	Number of Extracted Features		
	14	13	12
GFR	97.99	97.92	97.85
Kernel PCA	96.67	96.81	96.25
FastICA	95.63	95.28	92.85
UMAP	95.69	95.56	96.25
Autoencoder	95.14	94.44	94.03
LPP	90.56	90.35	89.03

TABLE III: Classification accuracy (%) of GFR with multilinear polynomials of degree up to 4, Kernel PCA, FastICA, UMAP, Autoencoder, and LPP over benchmark datasets.

$\{c_r Y_i Y_j\}_{i,j \in [n]}$ or $\{c_r Y_i Y_j Y_k\}_{i,j,k \in [n]}$, where c_r 's, $r = n+1, \dots, d$, are constant and chosen³ so that the variance reduction assumption (12) in Theorem 2 is satisfied. We apply GCA with $\mathcal{F}(\mathbf{z})$ being multilinear polynomials of degree up to 2 or 3, according to the degree of the redundancies mentioned above. We vary dataset sizes from 500 to 2000 and in each experiment GCA is applied, and its output is compared against the non-redundant principal components. We repeat the experiments for different dataset sizes 1000 times and the success rate of these experiments, expressed as a percentage, is reported in Table IV for two different types of redundancies. The results show that if the dataset size is large enough (roughly 1000 in these experiments), GCA can remove all redundant principal components from the data. However, as the dataset size decreases, the approximation of covariance matrices degrades. This degradation affects GCA's performance, particularly in datasets with redundancies of higher-degree multilinear polynomials.

Degree of Polynomials	Dataset Size					
	500	550	600	700	850	1000
2	93.6	96.2	97.6	99	99.8	100
3	68.6	84.5	96.5	97.9	99.5	100

TABLE IV: The success rate (%) of GCA in extracting correct non-redundant principal components in synthetic datasets with $d = 30$ features, among them $n = 15$ are mutually independent $Y_1, \dots, Y_{15}$, and the remaining 15 are redundancies of multilinear polynomials up to degree 2 or 3.

³For example, it is readily seen that $c_r < \sqrt{\min\{\text{Var}(Y_i), \text{Var}(Y_j)\} / (\text{Var}(Y_i) \text{Var}(Y_j))}$ suffices since Y_i and Y_j are mutually independent.

VIII. DISCUSSION AND CONCLUSIONS

This paper presents a generalization of the famous Principal Component Analysis (PCA). Unlike PCA, the presented methods, based on a Gram-Schmidt orthogonalization process over function spaces, can identify and remove nonlinear dependencies in the data, as long as such nonlinearities belong to the span of the chosen set of functions. We apply such techniques to perform feature extraction, showing an impressive ability to preserve significant amount of information with a much lower dimension, compared to PCA, and overall superior performance in classification accuracy w.r.t other state-of-the-art techniques on benchmark datasets.

REFERENCES

- [1] S. Khalid, T. Khalil, and S. Nasreen, "A survey of feature selection and feature extraction techniques in machine learning," in *2014 science and information conference*. IEEE, 2014, pp. 372–378.
- [2] K. Pearson, "Liili. on lines and planes of closest fit to systems of points in space," *The London, Edinburgh, and Dublin philosophical magazine and journal of science*, vol. 2, no. 11, pp. 559–572, 1901.
- [3] D. G. Luenberger, *Optimization by vector space methods*. John Wiley & Sons, 1997.
- [4] R. Vidal, Y. Ma, and S. Sastry, "Generalized principal component analysis (gpca)," *IEEE transactions on pattern analysis and machine intelligence*, vol. 27, no. 12, pp. 1945–1959, 2005.
- [5] E. J. Candès, X. Li, Y. Ma, and J. Wright, "Robust principal component analysis?" *Journal of the ACM (JACM)*, vol. 58, no. 3, pp. 1–37, 2011.
- [6] B. Schölkopf, A. Smola, and K.-R. Müller, "Kernel principal component analysis," in *Artificial Neural Networks—ICANN'97: 7th International Conference Lausanne, Switzerland, October 8–10, 1997 Proceedings*. Springer, 2005, pp. 583–588.
- [7] T. F. Cox and M. A. Cox, *Multidimensional scaling*. CRC press, 2000.
- [8] R. Larsen, "Decomposition using maximum autocorrelation factors," *Journal of Chemometrics: A Journal of the Chemometrics Society*, vol. 16, no. 8–10, pp. 427–435, 2002.
- [9] L. Wiskott and T. J. Sejnowski, "Slow feature analysis: Unsupervised learning of invariances," *Neural computation*, vol. 14, no. 4, pp. 715–770, 2002.
- [10] X. He and P. Niyogi, "Locality preserving projections," *Advances in neural information processing systems*, vol. 16, 2003.
- [11] A. Hyvärinen and E. Oja, "Independent component analysis: algorithms and applications," *Neural networks*, vol. 13, no. 4–5, pp. 411–430, 2000.
- [12] L. McInnes, J. Healy, and J. Melville, "Umap: Uniform manifold approximation and projection for dimension reduction," *arXiv preprint arXiv:1802.03426*, 2018.
- [13] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *science*, vol. 313, no. 5786, pp. 504–507, 2006.
- [14] K. D. Bollacker and J. Ghosh, "Linear feature extractors based on mutual information," in *Proceedings of 13th International Conference on Pattern Recognition*, vol. 2. IEEE, 1996, pp. 720–724.
- [15] J. M. Leiva-Murillo and A. Artes-Rodríguez, "Maximization of mutual information for supervised linear feature extraction," *IEEE Transactions on Neural Networks*, vol. 18, no. 5, pp. 1433–1441, 2007.
- [16] N. Kwak, "Feature extraction based on direct calculation of mutual information," *International Journal of Pattern Recognition and Artificial Intelligence*, vol. 21, no. 07, pp. 1213–1231, 2007.
- [17] K. Torkkola, "Feature extraction by non-parametric mutual information maximization," *Journal of machine learning research*, vol. 3, no. Mar, pp. 1415–1438, 2003.
- [18] A. Hyvärinen and E. Oja, "A fast fixed-point algorithm for independent component analysis," *Neural computation*, vol. 9, no. 7, pp. 1483–1492, 1997.
- [19] T. M. Cover, *Elements of information theory*. John Wiley & Sons, 1999.
- [20] D. Dua and C. Graff, "Uci machine learning repository," 2017. [Online]. Available: <http://archive.ics.uci.edu/ml>